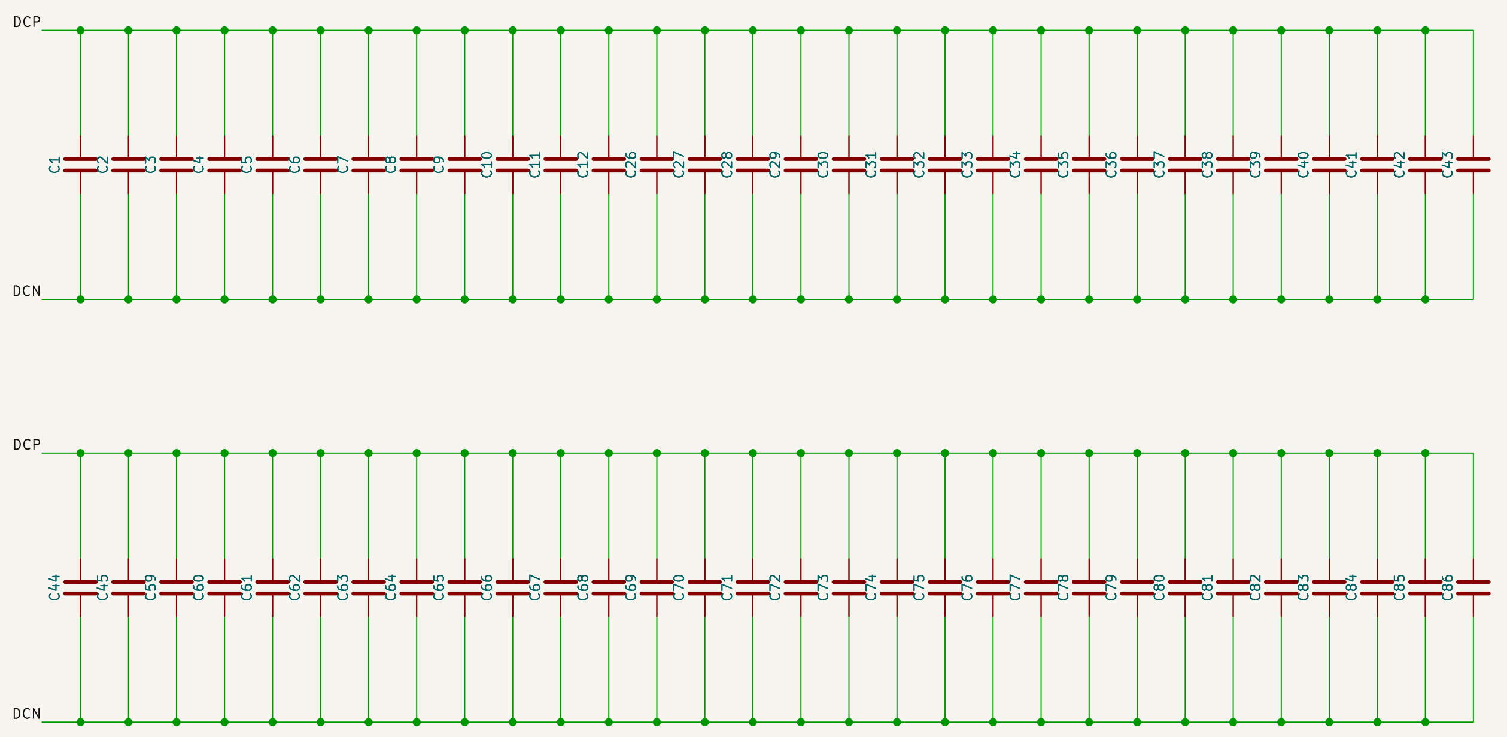
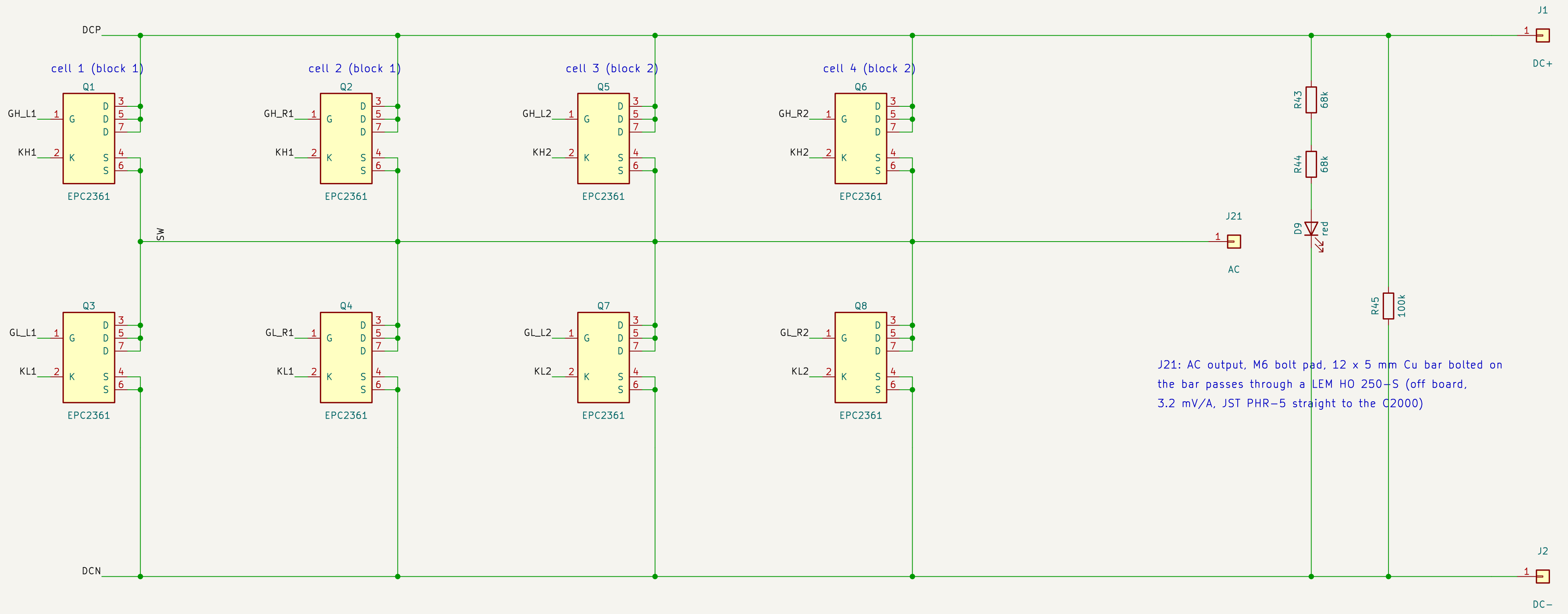
POWER STAGE AND DC LINK (block 1: Q1–Q4, block 2: Q5–Q8; one cell = HS + LS transistor, LS source straight to DC−; AC of all four cells joined on the PCB)


DCP
DCN
C44
C45
C59
C60
C61
C62
C63
C64
C65
C66
C67
C68
C69
C70
C71
C72
C73
C74
C75
C76
C77
C78
C79
C80
C81
C82
C83
C84
C85
C86

DC link, 60 capacitors: 40 x 1uF 100V X7S, 20 x 22uF 100V X7S 2220 (1 uF 0805: bypass at the HS drains of each cell, top + bottom; 22 uF 2220: bank)


DCP
cell 1 (block 1)
cell 2 (block 1)
cell 3 (block 2)
cell 4 (block 2)
Q1
Q2
Q5
Q6
GH_L1
GH_R1
GH_L2
GH_R2
KH1
KH2
G
K
D
S
EPC2361
SW
Q3
Q4
Q7
Q8
GL_L1
GL_R1
GL_L2
GL_R2
KL1
KL2
J1
DC+
J21
AC
J2
DC−
R43
68k
R44
68k
D9
red
R45
100k
DCN

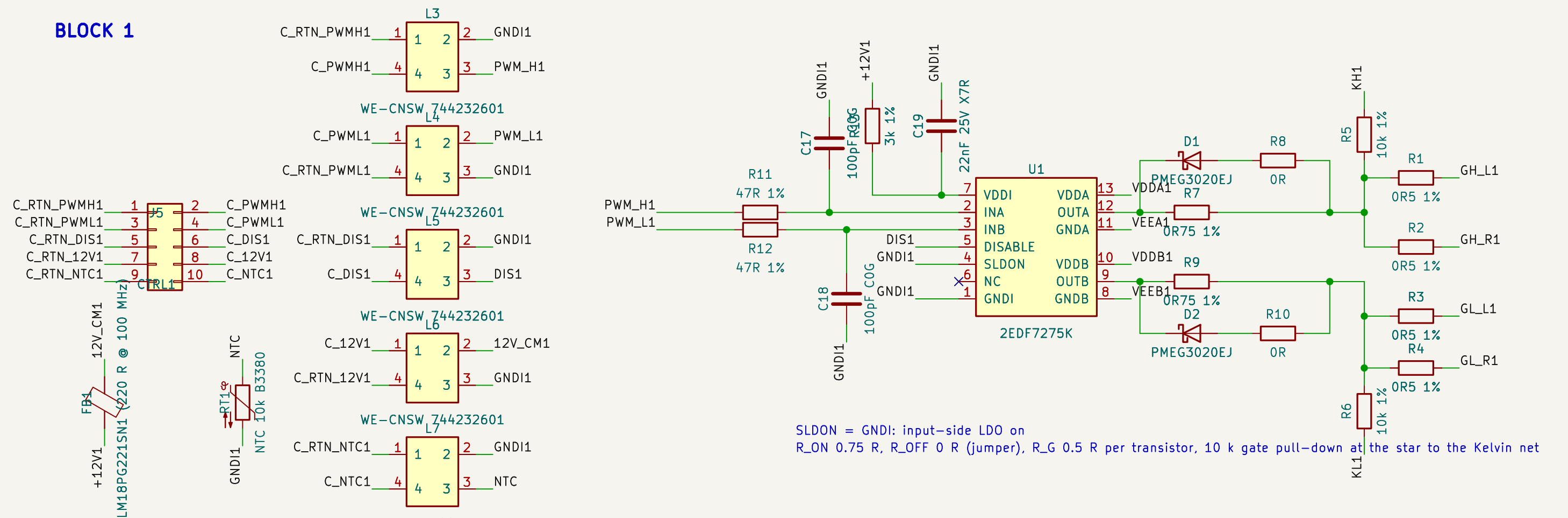
J21: AC output, M6 bolt pad, 12 x 5 mm Cu bar bolted on
the bar passes through a LEM HO 250–S (off board,
3.2 mV/A, JST PHR–5 straight to the C2000)

No shunt (load–test board): LS source stripes via–in–pad (blind L1–L2) into the DC− plane

DC link present LED, R45 100 k bleeder

GATE DRIVER BLOCKS: control header with common–mode chokes, 2EDF7275K input filters, R_ON / D + R_OFF, star, R_G per transistor (block 1 drives Q1–Q4, block 2 drives Q5–Q8)

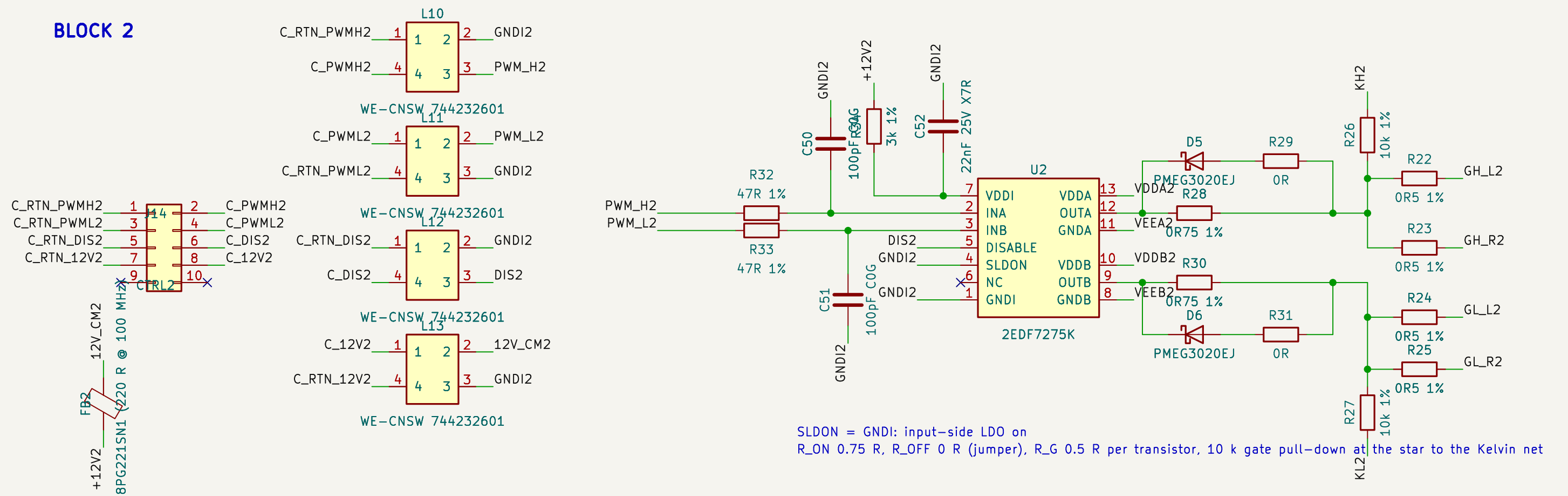
BLOCK 1


L3
C_RTN_PWMH1
C_PWMH1
GNDI1
PWM_H1
WE–CNSW 744232601
L4
C_PWML1
C_RTN_PWML1
PWM_L1
L5
C_RTN_DIS1
C_DIS1
DIS1
L6
C_12V1
C_RTN_12V1
12V_CM1
L7
C_RTN_NTC1
C_NTC1
NTC
J5
CTRL1
C_RTN_PWMH1
C_RTN_PWML1
C_RTN_DIS1
C_RTN_12V1
C_RTN_NTC1
C_PWMH1
C_PWML1
C_DIS1
C_12V1
C_NTC1
FB1
BLM18PG221SN1 (220 R @ 100 MHz)
12V_CM1
+12V1
RT1
NTC 10k B3380
GNDI1
PWM_H1
PWM_L1
R11
47R 1%
R12
47R 1%
C17
100pF C0G
R13
3k 1%
C18
100pF C0G
C19
22nF 25V X7R
+12V1
U1
2EDF7275K
VDDI
INA
INB
DISABLE
SLDON
NC
GNDI
VDDA
OUTA
GNDA
VDDB
OUTB
GNDB
VDDA1
VEEA1
VDDB1
VEEB1
D1
D2
PMEG3020EJ
R7
R9
0R75 1%
R8
R10
0R
R5
R6
10k 1%
KH1
KL1
R1
R2
R3
R4
0R5 1%
GH_L1
GH_R1
GL_L1
GL_R1

SLDON = GNDI: input–side LDO on
R_ON 0.75 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull–down at the star to the Kelvin net

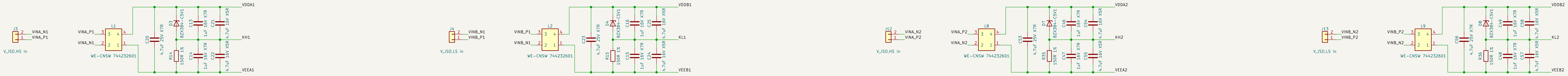
NTC on the control side (board temperature at the driver)

BLOCK 2


L10
C_RTN_PWMH2
C_PWMH2
GNDI2
PWM_H2
WE–CNSW 744232601
L11
C_PWML2
C_RTN_PWML2
PWM_L2
L12
C_RTN_DIS2
C_DIS2
DIS2
L13
C_12V2
C_RTN_12V2
12V_CM2
J14
C_RTN_PWMH2
C_RTN_PWML2
C_RTN_DIS2
C_RTN_12V2
C_PWMH2
C_PWML2
C_DIS2
C_12V2
FB2
BLM18PG221SN1 (220 R @ 100 MHz)
+12V2
PWM_H2
PWM_L2
R32
47R 1%
R33
47R 1%
C50
100pF C0G
C51
100pF C0G
C52
22nF 25V X7R
R34
3k 1%
U2
2EDF7275K
VDDA2
VEEA2
VDDB2
VEEB2
D5
D6
PMEG3020EJ
R28
R30
0R75 1%
R29
R31
0R
R26
R27
10k 1%
KH2
KL2
R22
R23
R24
R25
0R5 1%
GH_L2
GH_R2
GL_L2
GL_R2

SLDON = GNDI: input–side LDO on
R_ON 0.75 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull–down at the star to the Kelvin net

GATE SUPPLY ISLANDS: external isolated supply (one HS, one LS, split per block) –> WE–CNSW –> 5.1 V Zener VDD–S, 150 R bias S–VEE: V_GS,on = +5.1 V, V_GS,off = –(V_ISO – 5.1 V)


J3
VINA_N1
VINA_P1
V_ISO,HS in
L1
WE–CNSW 744232601
C20
4.7uF 25V X7R
D3
BZX384–C5V1
R14
150R 1%
C13
C14
1uF 16V X7R
C21
C22
4.7uF 16V X5R
VDDA1
KH1
VEEA1
J4
VINB_N1
VINB_P1
V_ISO,LS in
L2
C23
D4
R15
C16
C24
C25
VDDB1
KL1
VEEB1
J12
VINA_N2
VINA_P2
L8
C53
D7
R35
C46
C47
C54
C55
VDDA2
KH2
VEEA2
J13
VINB_N2
VINB_P2
L9
C56
D8
R36
C48
C49
C57
C58
VDDB2
KL2
VEEB2

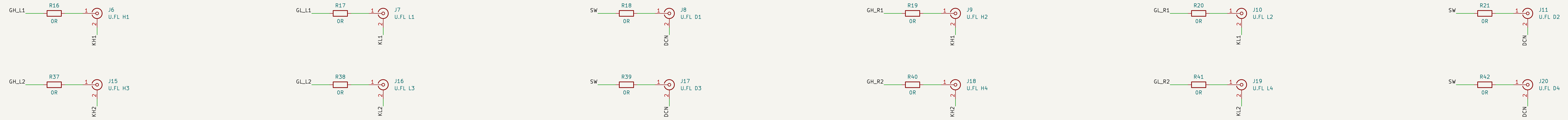
J3 / L1: HS block 1

J4 / L2: LS block 1

J12 / L8: HS block 2

J13 / L9: LS block 2

U.FL MEASUREMENT PORTS (bottom side, at each cell): V_GS HS / LS (0 R, shield = Kelvin), V_DS LS (4.99 k 2512 series into the 50 R scope input = 100:1, 1.1 W at 75 V; shield = LS source = DC–)


GH_L1
R16
0R
J6
U.FL H1
KH1
GL_L1
R17
J7
U.FL L1
KL1
SW
R18
J8
U.FL D1
DCN
GH_R1
R19
J9
U.FL H2
GL_R1
R20
J10
U.FL L2
SW
R21
J11
U.FL D2
GH_L2
R37
J15
U.FL H3
KH2
GL_L2
R38
J16
U.FL L3
KL2
SW
R39
J17
U.FL D3
GH_R2
R40
J18
U.FL H4
GL_R2
R41
J19
U.FL L4
SW
R42
J20
U.FL D4

MOUNTING: H1–H4 board corners (insulated), H5–H7 heatsink spreader; heatsink HS1 (Fischer LAM 4, 100 mm, rotated 90 deg, tube across the board width) is a board–only item

H1
H2
H3
H4
H5
H6
H7
M2.5

Nets: DCP = DC+, DCN = DC–. SW = joined switch node of all four cells. KH1/KH2 and KL1/KL2 are the Kelvin source nets (pad 2 of each EPC2361); they meet the power source only inside the package.

LS power source = DC– directly (no shunt). The LS gate drive and the LS supply are referenced to the Kelvin net KLx, not to DC–.

Block suffix 1/2 on every driver–side net (GNDI1, VDDA2, ...). Control side (CTRLx 2 x 5 header): PWM_H, PWM_L, DIS, 12 V and (block 1) NTC each with their own return C_RTN_* through one WE–CNSW common–mode choke per pair; 12 V then through a ferrite bead. The control side is a top–layer–only area with a 1200 V barrier (4 mm outer, 3 mm L2) to all power copper; see reports/pcb–rules/PCB_rules_and_checklist.md.

No short–circuit protection on this board. Closed–loop control from a TI TMS320F28379D over the CTRL headers.

generated by scripts/make_kicad_sch_revb.py; netlist identical to the PCB (check_sch_vs_pcb.py)
4 x EPC2361 per switch, 2 x 2EDF7275K, no shunt, LAM 4 heatsink, external isolated supplies, U.FL p
KTH
Sheet: /
File: spb–bb–revb–load.kicad_sch
Title: SPB leg, Rev B, load–test board
Size: A1
Date: 2026–10–14
Rev: B
KiCad E.D.A. kicad–cli 7.0.10
Id: 1/1